

Software Requirements Specification (SRS) for Lanezy

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1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for Lanezy, an AI-driven traffic management system.

1.2 Scope

Lanezy is a software solution that integrates with physical infrastructure (traffic cameras and IoT sensors).

In Scope:

- Ingestion of real-time video feeds and sensor data.
- Traffic density estimation using Computer Vision.
- Adaptive signal timing using Reinforcement Learning.
- Web dashboard for traffic authorities.
- Simulation to validate a 10% reduction in commute time.

Out of Scope:

- Installation of physical hardware.
- Enforcement of traffic laws unless added later.

1.3 Definitions and Acronyms

TMS: Traffic Management System

RL: Reinforcement Learning

IoT: Internet of Things

CV: Computer Vision

Admin: Traffic Control Authority User

2. Overall Description

2.1 Product Perspective

Lanezy functions as a centralized intelligence layer on top of existing traffic infrastructure.

2.2 User Characteristics

- Traffic Engineers
- Emergency Services
- General Commuters

2.3 Assumptions and Dependencies

- High-speed internet at intersections
- Cameras provide adequate resolution
- Signal controllers accept external timing commands

3. System Architecture

- Data Ingestion Layer
- Processing Layer (Vision + RL Engine)
- Control Layer
- Presentation Layer

4. Functional Requirements

4.1 Real-Time Traffic Analysis

REQ■1: Accept video streams via RTSP/RTMP

REQ■2: Detect vehicles with >90% accuracy

REQ■3: Compute traffic density and queue length

4.2 Adaptive Signal Control

REQ■4: RL model sets optimal green duration

REQ■5: Update signal timing per cycle

REQ■6: Prevent starvation using max wait limit

4.3 Administrative Dashboard

REQ■7: Live city map with congestion indicators

REQ■8: View live video

REQ■9: Provide historical analytics

4.4 Manual Override & Safety

REQ■10: Manual Override button

REQ■11: Fail■safe fallback to pre■timed cycles

5. Non-Functional Requirements

Performance:

- Latency < 2 seconds
- Handle 50 video streams per node

Reliability:

- 99.9% uptime
- Recovery < 30 seconds

Security:

- MFA authentication

- TLS 1.3 encryption

Scalability:

- Add new intersections without restart

6. External Interface Requirements

User Interfaces:

- Web dashboard
- Mobile alerts

Hardware Interfaces:

- IP cameras
- Traffic controller protocols (NTCIP 1202)

7. Data Models

7.1 Intersection Entity

ID, Location, Lanes, Current State

7.2 Traffic Log Entity

Timestamp, Intersection ID, Vehicle Count, Queue Length, Action Taken

8. Simulation & Validation Strategy

- Baseline simulation with static timers
- Lanezy simulation
- Require $\geq 10\%$ reduction in commute time